Lewis acids	Lewis bases
Electron deficient molecules such as BF_3,$AlCl_3$,BeF_2 etc...	Molecules with one (or) more lone pairs of electrons. NH_3,H_2O,R-O-H,R-O-R, R - NH_2
All metal ions Examples: Fe^{2+},Fe^{3+},Cr^{3+},Cu^{2+} etc...	All anions F^-,Cl^-,CN^-,SCN^-,SO_4^{2-} etc...
Molecules that contain a polar double bond Examples : SO_2,CO_2,SO_3 etc...	Molecules that contain carbon – carbon multiple bond Examples: $CH_2{=}CH_2$,$CH \equiv CH$ etc...
Molecules in which the central atom can expand its octet due to the availability of empty d – orbitals Example: SiF_4,SF_4,$FeCl_3$ etc..	All metal oxides CaO,MgO,Na_2O etc...
Carbonium ion $(CH_3)_3\,C^+$	Carbanion CH_3^-

Example

Identify the Lewis acid and the Lewis base in the following reactions.

$Cr^{3+} + 6\ H_2O \rightarrow [Cr(H_2O)_6]^{3+}$

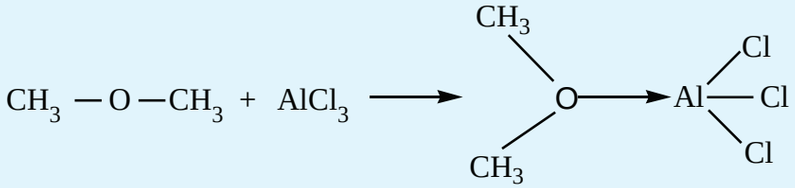
In the hydration of ion, each of six water molecules donates a pair of electron to Cr^{3+} to from the hydrated cation, hexaaquachromium (III) ion, thus, the Lewis acid is Cr^{3+} and the Lewis base H_2O.

Evaluate yourself – 3

Identify the Lewis acid and the Lewis base in the following reactions.

i. $CaO + CO_2 \rightarrow CaCO_3$

ii.



XII U8-Ionic equilibrium.indd 5 11-01-2021 12:58:15